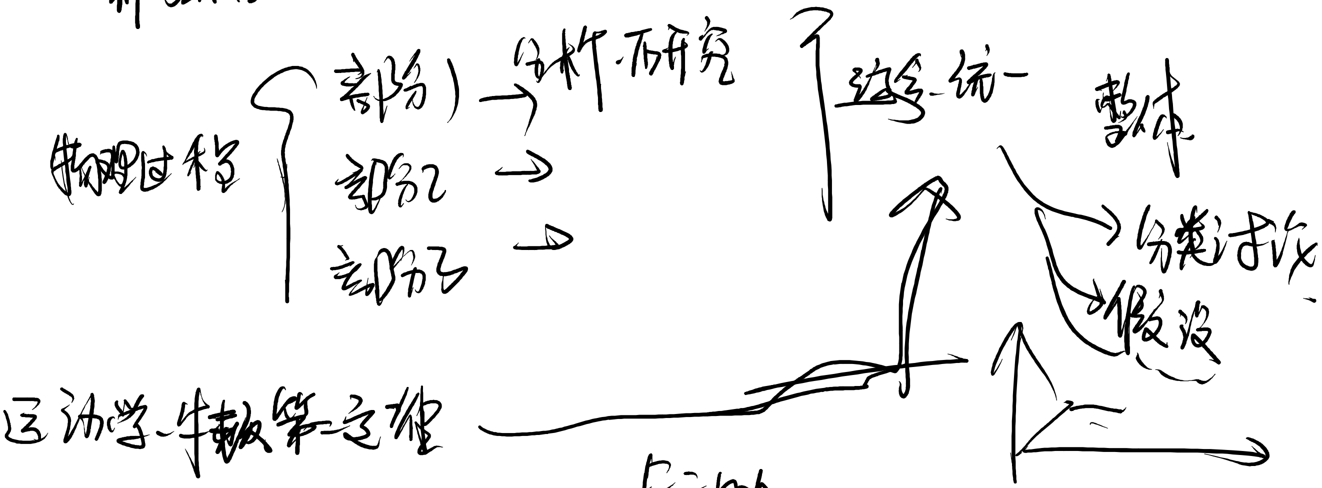
Chapter 1 分析与综合。

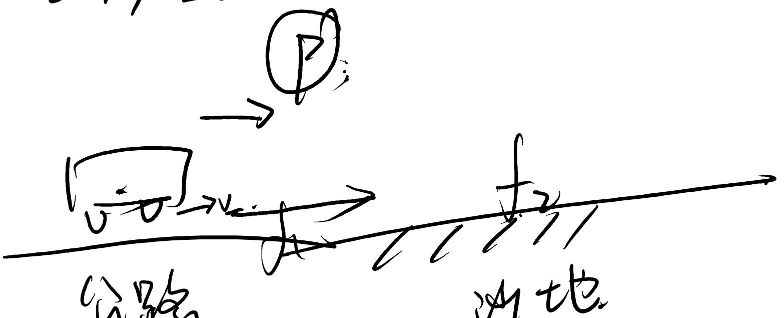


$F=ma$

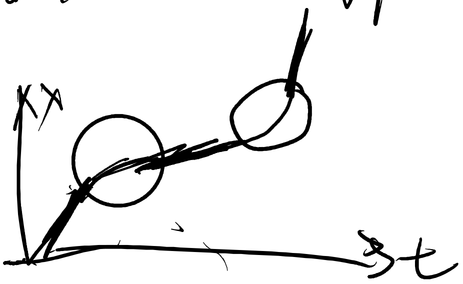
$v=v_0+at$

$x=v_0t+\frac{1}{2}at^2$

2013 上海



$f_2>f_1$



① 公路匀速

② 公路→沙地

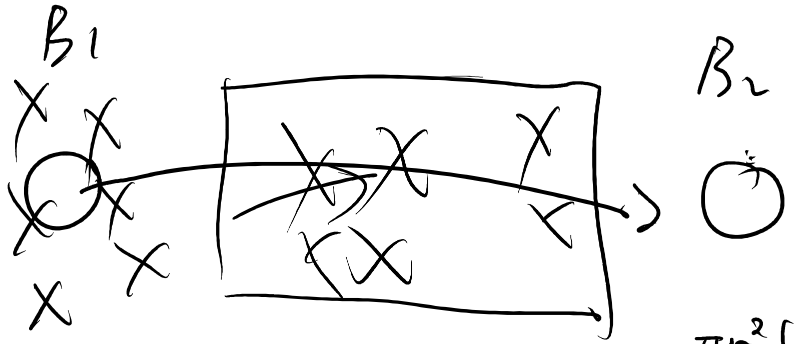
③ 沙地匀速

④

⑤

$ma+f=F_{牵}=\frac{P}{v}$

$a=\frac{P}{mv}-\frac{f}{m}$ $\quad \frac{P}{v}-f$

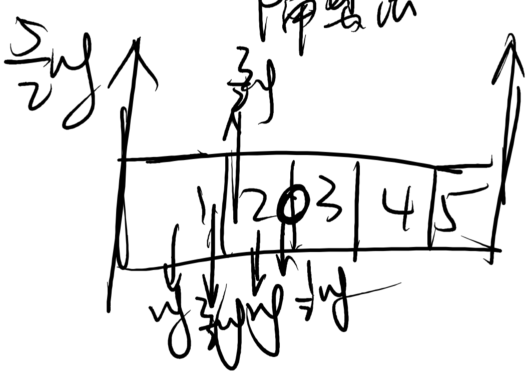


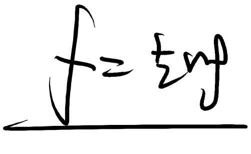
$$q=\frac{\Delta\Phi}{R}=\frac{\pi r^2(B_2-B_1)}{R}$$

研究对象的分析综合 → 整体

↓

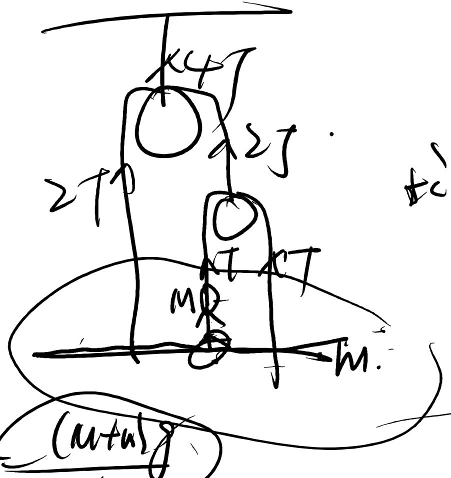
隔离法





(2)

$$T=\frac{(M+m)g}{2}$$



求 T、N

↑a

$$4T=\frac{(M+m)g}{4}$$

$$N_2=\frac{(3M-m)}{4}g$$

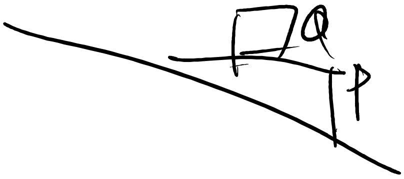
T、N

T. N　↓$(M+m)g$

$4T-(M+m)g=(M+m)a.$

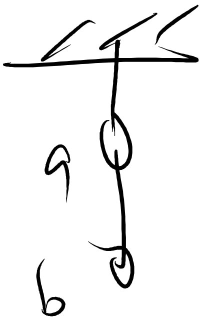
$$N=\frac{(3M-m)(g+a)}{4}$$

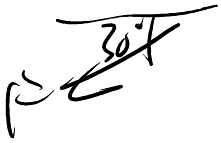
2013 广东



P 滑还是不滑. $(m_Q>m_P\tan\theta)$

$(m_Q<m_P\tan\theta).$

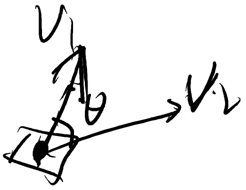


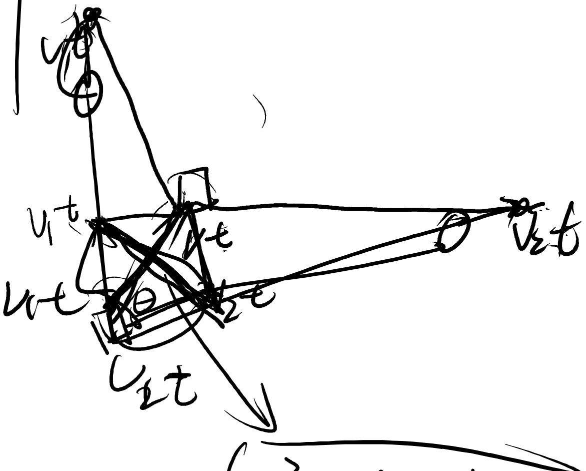


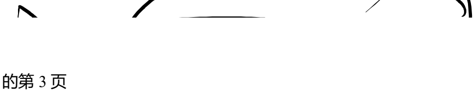
微元分析法

微元. 微小过程. 曲→直线

变→常





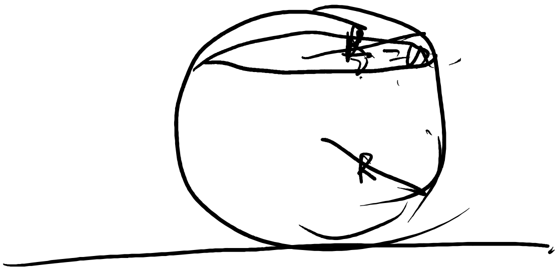


$t\sqrt{v_1^2+v_2^2-2v_1v_2\cos\theta}$

$$t' = \frac{\sqrt{v_1^2+v_2^2-2v_1v_2\cos\theta}}{\sin\theta} \cdot t$$

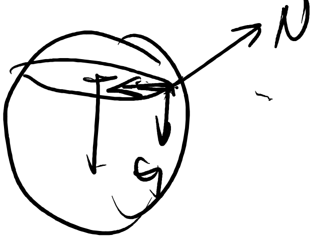
$t\sqrt{v_1^2+v_2^2-2v_1v_2\cos\theta}$

$v = t$

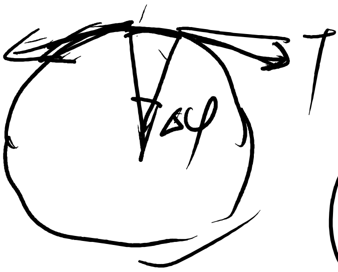


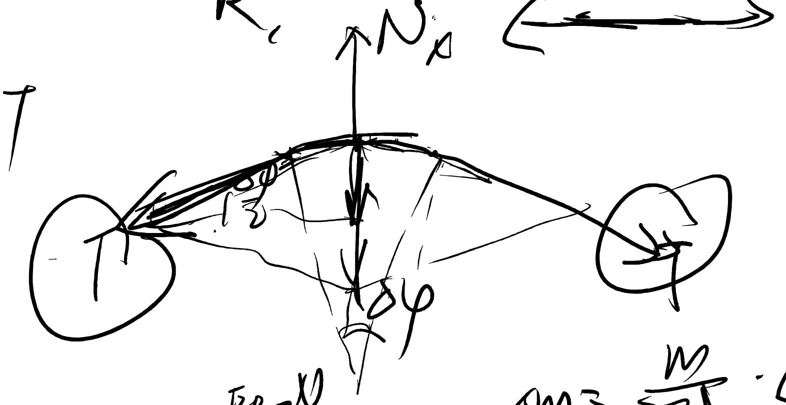
$2\pi a \qquad a = \frac{R}{2}$

$2\pi b \qquad b = \sqrt{2}\,a$



$k,$

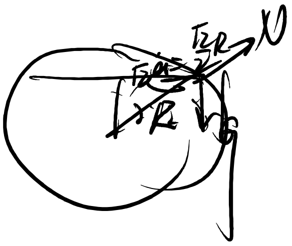




$\lambda = \frac{m}{2\pi b}$

$\Delta\varphi \to 0$

$\Delta m = \frac{m}{2\pi b}\cdot\Delta\varphi\, b$



$\sin\Delta\varphi \approx \Delta\varphi$

$$2T\sin\frac{\Delta\varphi}{2} = \frac{m\Delta\varphi}{2\pi}g$$

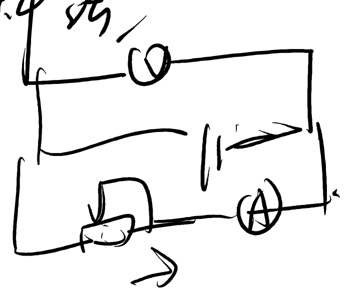
$$2T\cdot\frac{\Delta\varphi}{2} = \frac{m\Delta\varphi g}{2\pi}$$

$$T = \frac{mg}{2\pi}$$

$$K = \frac{T}{2\pi(b-a)}$$

定性分析法

定性分析法



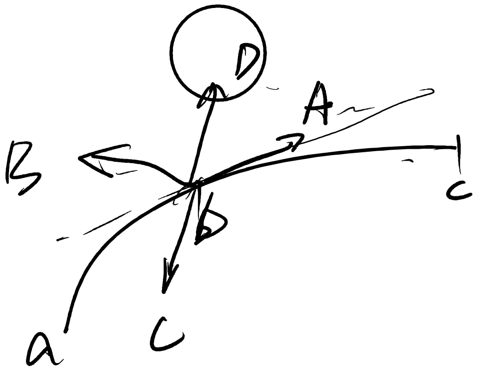
A. P总 ↓

B. U ↑

C. U损 ↓

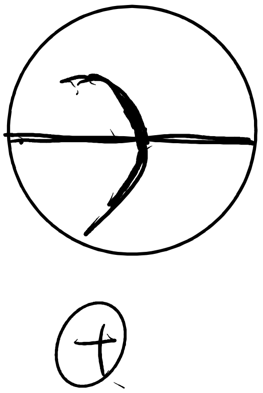
D. P输 ↑↓

ABC



e^-

v↓



× × ×

v↓

×

F ×